

CBSE / NCERT | CLASS X | 2026-27 PRACTICE EDITION

Mathematics

Assertion and Reason

All textbook chapters, with scope labels

15 chapters | 30 questions | answer key with explanations

How to answer

Judge Assertion (A) and Reason (R) separately, then decide whether R explains A.

A	Both A and R are true, and R explains A.
B	Both A and R are true, but R does not explain A.
C	A is true, but R is false.
D	A is false, but R is true.

The chapter set includes textbook extension chapters too. Status labels distinguish current board content, formative/project/map-only items and chapters not listed in the cited annual syllabus.

01. Real Numbers

Current course

Q1

Assertion (A): The square root of 2 is irrational.

Reason (R): If it were p/q in lowest terms, squaring would force both p and q to be even.

Q2

Assertion (A): For positive integers a and b , $HCF(a,b) \times LCM(a,b) = a \times b$.

Reason (R): For every prime, the exponents in the HCF and LCM are the minimum and maximum of the two exponents.

02. Polynomials

Current course

Q1

Assertion (A): The zeros of $x^2 - 5x + 6$ are 2 and 3.

Reason (R): For ax^2+bx+c , the sum and product of zeros are $-b/a$ and c/a .

Q2

Assertion (A): A non-zero polynomial of degree 3 can have at most three distinct zeros.

Reason (R): Each distinct zero gives a different linear factor of the polynomial.

03. Pair of Linear Equations in Two Variables

Current course

Q1

Assertion (A): The equations $2x+3y=6$ and $4x+6y=12$ have infinitely many solutions.

Reason (R): The second equation is exactly twice the first equation.

Q2

Assertion (A): Two non-parallel lines in a plane have exactly one point of intersection.

Reason (R): A unique common point gives the unique solution of the corresponding pair of equations.

04. Quadratic Equations

Current course

Q1

Assertion (A): A quadratic equation with discriminant zero has equal real roots.

Reason (R): The quadratic formula contains the term square root of the discriminant.

Q2**Assertion (A):** The equation $x^2+4x+5=0$ has no real roots.**Reason (R):** Its discriminant is $16-20$, which is negative.

05. Arithmetic Progressions

Current course

Q1**Assertion (A):** The sequence 7, 11, 15, 19 is an arithmetic progression.**Reason (R):** The difference between each pair of consecutive terms is 4.**Q2****Assertion (A):** The sum of the first 10 terms of 2, 5, 8, ... is 155.**Reason (R):** The sum of n AP terms is $n/2$ times the sum of the first and last terms.

06. Triangles

Current course

Q1**Assertion (A):** A line parallel to one side of a triangle divides the other two sides in the same ratio.**Reason (R):** The small triangle formed is similar to the original triangle.**Q2****Assertion (A):** If two angles of one triangle equal two angles of another, the triangles are similar.**Reason (R):** The third angles must also be equal because each triangle's angles sum to 180 degrees.

07. Coordinate Geometry

Current course

Q1**Assertion (A):** The distance between (0,0) and (3,4) is 5 units.**Reason (R):** The distance formula follows from the Pythagorean theorem on the coordinate plane.**Q2****Assertion (A):** The midpoint of (2,4) and (8,10) is (5,7).**Reason (R):** The midpoint coordinates are the averages of the corresponding endpoint coordinates.

08. Introduction to Trigonometry

Current course

Q1**Assertion (A):** For an acute angle θ in a right triangle, $\sin(\theta) = \text{opposite/hypotenuse}$.**Reason (R):** Similar right triangles with the same acute angle have the same side ratios.**Q2****Assertion (A):** $\sin(30^\circ) = 1/2$.**Reason (R):** $\cos(60^\circ) = 1/2$.

09. Some Applications of Trigonometry

Current course

Q1**Assertion (A):** For a fixed tower, a larger angle of elevation corresponds to a shorter horizontal distance from its base.**Reason (R):** $\tan(\theta) = \text{tower height} / \text{horizontal distance}$.**Q2****Assertion (A):** The angle of depression from an observer equals the angle of elevation from the object below.**Reason (R):** The two horizontal lines are parallel, so the sight line forms alternate interior angles.

10. Constructions

Textbook extension; not listed in the current CBSE course structure

Q1**Assertion (A):** A perpendicular bisector divides a line segment into two equal parts at 90° .**Reason (R):** Every point on the perpendicular bisector is equidistant from the segment's endpoints.**Q2****Assertion (A):** A ray divided in the ratio 2:3 has five equal parts in total.**Reason (R):** The two ratio numbers count the equal divisions assigned to each side.

11. Circles

Current course

Q1**Assertion (A):** A tangent to a circle is perpendicular to the radius at the point of contact.**Reason (R):** The perpendicular from a centre to a line gives the shortest distance to that line.

Q2**Assertion (A):** Tangents drawn from the same external point to a circle have equal lengths.**Reason (R):** The radii to the contact points are equal and each tangent is perpendicular to its radius.

12. Areas Related to Circles

Current course

Q1**Assertion (A):** A 90-degree sector has one-quarter the area of its circle.**Reason (R):** For a fixed radius, sector area is proportional to its central angle.**Q2****Assertion (A):** The area of a minor segment equals the sector area minus the area of its triangle.**Reason (R):** A segment is the region between a chord and its corresponding arc.

13. Surface Areas and Volumes

Current course

Q1**Assertion (A):** When a solid is melted and recast without loss, its volume remains the same.**Reason (R):** The volume of a sphere is $\frac{4}{3} \pi r^3$.**Q2****Assertion (A):** Joining two solids can change their exposed surface area without changing their combined volume.**Reason (R):** The joined faces become internal and are no longer exposed to the outside.

14. Statistics

Current course

Q1**Assertion (A):** In the direct method, the mean of grouped data is $\frac{\sum f_i x_i}{\sum f_i}$.**Reason (R):** Each class mark represents the observations in its class interval.**Q2****Assertion (A):** For a symmetric unimodal distribution, mean, median and mode can coincide.**Reason (R):** Symmetry places the balance point and middle observation at the centre of the distribution.

15. Probability

Current course

Q1**Assertion (A):** The probability of getting a head on a fair coin toss is $\frac{1}{2}$.**Reason (R):** The two outcomes, head and tail, are equally likely.**Q2****Assertion (A):** For any event E, $P(E)$ lies between 0 and 1 inclusive.**Reason (R):** The number of favourable outcomes cannot be negative or exceed the total outcomes.

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Answer key and explanations

01. Real Numbers

Q1 - A. The assumed fraction contradicts lowest terms, proving irrationality.

Q2 - A. The minimum and maximum exponents add to the original total exponent.

02. Polynomials

Q1 - A. The roots 2 and 3 have sum 5 and product 6, matching the coefficients.

Q2 - A. Four distinct zeros would require at least four linear factors, exceeding degree 3.

03. Pair of Linear Equations in Two Variables

Q1 - A. They represent the same line, so every point on it satisfies both.

Q2 - A. The intersection point is precisely the ordered pair satisfying both equations.

04. Quadratic Equations

Q1 - A. When the discriminant is zero, both signs in the formula give the same root.

Q2 - A. A negative discriminant makes the square root in the real quadratic formula impossible.

05. Arithmetic Progressions

Q1 - A. A constant consecutive difference is the defining property of an AP.

Q2 - A. The tenth term is 29, so the sum is $10/2 \times (2+29) = 155$.

06. Triangles

Q1 - A. Similarity gives proportional corresponding sides, which yields the equal ratios.

Q2 - A. All corresponding angles are equal, satisfying the AA similarity criterion.

07. Coordinate Geometry

Q1 - A. The horizontal and vertical differences are 3 and 4, giving distance 5.

Q2 - A. A point halfway along a segment is halfway in both coordinate directions.

08. Introduction to Trigonometry

Q1 - A. Similarity makes the trigonometric ratio independent of the triangle's size.

Q2 - B. Both standard values are true, but the second does not explain the first.

09. Some Applications of Trigonometry

Q1 - A. With fixed height, increasing $\tan(\theta)$ requires the denominator to decrease.

Q2 - A. The parallel-line angle relationship makes the two angles equal.

10. Constructions

Q1 - A. Its intersection with the segment is the midpoint, and the constructed line is perpendicular.

Q2 - A. Two parts plus three parts make five equal intervals.

11. Circles

Q1 - A. A tangent touches at one point only, so the centre's shortest radius reaches that contact perpendicularly.

Q2 - A. The resulting right triangles share the external-point hypotenuse and have equal radii, so they are congruent.

12. Areas Related to Circles

Q1 - A. Ninety degrees is one-quarter of a full 360-degree turn.

Q2 - A. Removing the triangle below the chord from the sector leaves the segment.

13. Surface Areas and Volumes

Q1 - B. The formula gives a sphere's volume, while conservation follows from the recasting process.

Q2 - A. Internal faces stop contributing to exposed surface area, while the material volume is still added.

14. Statistics

Q1 - A. Multiplying each class mark by its frequency weights it by the number of observations.

Q2 - A. For a symmetric single-peaked distribution, the centre is also the peak and midpoint.

15. Probability

Q1 - A. One of two equally likely outcomes is a head.

Q2 - A. Dividing a count in that range by the total count gives a value from 0 to 1.

Scope and sources

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Question text in this workbook is original practice material. Chapter labels and scope notes are provided for navigation; check your school's current list before using extension chapters for exam revision.

CBSE Class X Mathematics curriculum 2026-27

https://cbseacademic.nic.in/web_material/CurriculumMain27/SecPart1/Maths_SecP1X_2026-27.pdf

CBSE curriculum portal

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